Project: Worked examples
Title: Duct sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Sizes each duct section by the equal-friction method (a target friction rate) or the velocity method (a target velocity), rounds to standard sizes, and reports velocity, friction rate, friction and fitting losses per section and the total pressure drop along the index run, with warnings against user velocity and pressure-drop limits.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Round duct diameter from the design target | D by Colebrook with the air properties at T (equal friction), or D = √(4Q/πv) (velocity), rounded up to the next standard size |
| Rectangular duct of equal friction and capacity | $D_e$ = 1.30 (ab)^0.625/(a + b)^0.25 |
| Flat-oval duct of equal friction and capacity | $D_e$ = 1.55 A^0.625/P^0.25, with A = $\pi b^2/4$ + b(a − b) and P = πb + 2(a − b) |
| Velocity method, rectangular or flat-oval duct with side a fixed | b from the free area A = Q/v (rectangle b = A/a; flat oval the smaller root of $(1 - \pi/4)b^2 - ab + A = 0$), rounded up; $D_e$ is kept for the friction only |
| Friction loss per metre, at the velocity $v_e$ of the equivalent round duct | $\Delta p/L = f\,\rho\,v_e^2/(2D_e)$ |
| Fitting losses, at the actual velocity | $\Delta p = \Sigma C\,\rho\,v^2/2$ |
| Index-run totals, summed over every section of the run | $\Delta p_{run} = \Sigma(\Delta p/L \cdot L + \Sigma C\,\rho\,v^2/2)$ |

Basis: Duct design (friction chart basis, equivalent diameters, loss coefficients)[1](Ch. 21); Flow of fluids in pipes and ducts[2](Section 4); Ventilation for buildings: sheet metal air ducts and fittings with rectangular cross-section, dimensions[3]; Ventilation for buildings: sheet metal air ducts and fittings with circular cross-section, dimensions[4].

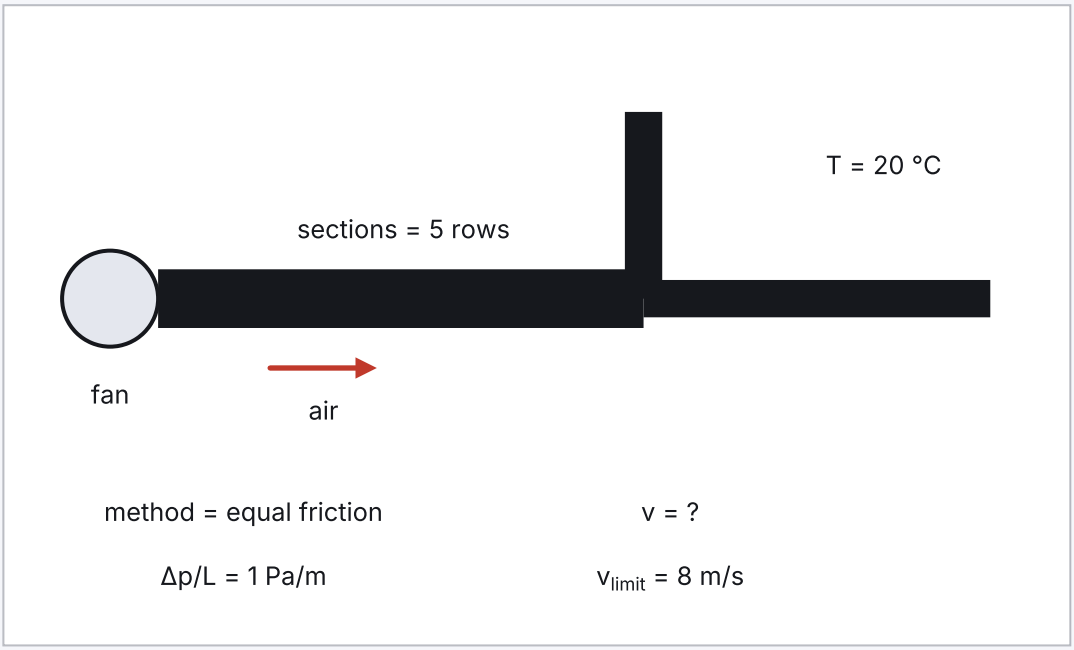



Figure 1 — A duct run from the fan through its sections as listed (a main and two branches drawn), each sized by the chosen method. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| method | Sizing method | equal friction | | assumption: Equal friction for a low-pressure system. | any |
| $dp_{target}$ | Target friction rate (equal friction) | 1 | Pa/m | spec: SPEC-M-07 §5: 1 Pa/m design friction rate | 0.3 … 5 Pa/m |
| $v_{target}$ | Target velocity (velocity method) | 6 | m/s | default | 2 … 20 m/s |
| T | Air temperature | 20 | °C | assumption: Supply air 20 °C. | -20 … 60 °C |
| $p_{atm}$ | Barometric pressure | 101.325 | kPa | assumption: Sea level. | 80 … 105 kPa |
| eps | Duct roughness | 0.09 | mm | spec: Galvanised sheet, medium roughness (ASHRAE Ch. 21 category) | 0.03 … 3 mm |
| sections | Duct sections | 5 row(s) | | drawing: DRG-M-205 rev B | any |
| $v_{limit}$ | Velocity limit (warning and check) | 8 | m/s | spec: SPEC-M-07 §5: 8 m/s in risers | 2 … 30 m/s |
| $dp_{limit}$ | Total pressure-drop limit (check) | 250 | Pa | spec: Fan external static allowance 250 Pa | 50 … 2000 Pa |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $round_{to}$ | Round sizes up to a multiple of | 50 | mm | spec: BS EN 1505 / 1506 preferred sizes, 50 mm steps | 1 … 100 mm |

| TABLE ROW | VALUES |
|---|---|
| sections[1] | name = AHU to riser; Q = 2 m³/s; L = 15 m; shape = rectangular; a = 800 mm; D = —; C = 0.8; dp_eq = 0 Pa; index = yes |
| sections[2] | name = riser; Q = 2 m³/s; L = 12 m; shape = round; a = —; D = —; C = 0.3; dp_eq = 0 Pa; index = yes |
| sections[3] | name = floor main; Q = 1.2 m³/s; L = 25 m; shape = rectangular; a = 500 mm; D = —; C = 1.1; dp_eq = 0 Pa; index = yes |
| sections[4] | name = branch to VAV; Q = 0.4 m³/s; L = 10 m; shape = round; a = —; D = —; C = 0.9; dp_eq = 60 Pa; index = yes |
| sections[5] | name = other branch; Q = 0.8 m³/s; L = 18 m; shape = round; a = —; D = —; C = 1; dp_eq = 60 Pa; index = no |

## 3. Assumptions

A1. Air is dry at the stated temperature and pressure (ρ = p/(287.055 T)); viscosity by Sutherland's law. (Standard air properties.)
A2. Rectangular and flat-oval ducts are sized through the equivalent diameter for equal friction and capacity. (Huebscher / Heyt–Diaz relations as given in ASHRAE Fundamentals Ch. 21.)
A3. Fitting loss coefficients C are user-supplied (manufacturer or ASHRAE/CIBSE data, not embedded) and refer to the section's velocity pressure. (Non-negotiable: no standard tables embedded.)
A4. method (Sizing method) is taken as equal friction. (Equal friction for a low-pressure system.)
A5. T (Air temperature) is taken as 20 °C. (Supply air 20 °C.)
A6. $p_{atm}$ (Barometric pressure) is taken as 101.325 kPa. (Sea level.)

## 4. Working

[10] ρ = p / (287.055 T)
= 101320 Pa / (287.055 × 293.15 K)
= 1.2041 kg/m³
Air density

[11] μ = Sutherland's law
= T = 293.15 K
= 18.133 uPa·s
Air dynamic viscosity

[12] $a_{s1}$ = AHU to riser: side a
= as given
= 800 mm
Side a, AHU to riser

[13] $b_{s1}$ = AHU to riser: side b
= from the target equivalent diameter 573.91 mm
= 400 mm
Side b, AHU to riser

[14] $De_{s1}$ = AHU to riser: equivalent diameter [1](Ch. 21)
= 1.30 (ab)^0.625/(a + b)^0.25
= 609.35 mm
Equivalent diameter, AHU to riser

[15] $v_{s1}$ = AHU to riser: Q / A
= 2 / 0.32
= 6.25 m/s
Velocity, AHU to riser
[16] $Re_{s1}$ = AHU to riser: ρ $v_e$ $D_e$ / μ ($v_e$ = 4Q/π$D_e^2$)
= 1.2041 × 6.8581 × 0.60935 / 0.000018133
= 277500
Reynolds number, AHU to riser
[17] $f_{s1}$ = AHU to riser: friction factor
= Colebrook–White
= 0.015999
Friction factor, AHU to riser
[18] $dpL_{s1}$ = AHU to riser: f ρ $v_e^2$ / (2 $D_e$)
= 0.015999 × 1.2041 × 6.8581² / (2 × 0.60935)
= 0.74347 Pa/m
Friction rate, AHU to riser
[19] $dp_{f,s1}$ = AHU to riser: (Δp/L) L
= 0.74347 × 15
= 11.152 Pa
Friction loss, AHU to riser
[20] $dp_{fit,s1}$ = AHU to riser: ΣC ρ v²/2
= 0.8 × 23.518
= 18.814 Pa
Fitting loss, AHU to riser
[21] $dp_{s1}$ = AHU to riser: friction + fittings + equipment
= 11.152 + 18.814 + 0
= 29.966 Pa
Section loss, AHU to riser (index run)
[22] $D_{s2}$ = riser: diameter
= target 573.91 mm rounded up to 50 mm
= 600 mm
Diameter, riser
[23] $v_{s2}$ = riser: Q / A
= 2 / 0.28274
= 7.0736 m/s
Velocity, riser
[24] $Re_{s2}$ = riser: ρ $v_e$ $D_e$ / μ ($v_e$ = 4Q/π$D_e^2$)
= 1.2041 × 7.0736 × 0.6 / 0.000018133
= 281820
Reynolds number, riser
[25] $f_{s2}$ = riser: friction factor
= Colebrook–White
= 0.015985
Friction factor, riser
[26] $dpL_{s2}$ = riser: f ρ $v_e^2$ / (2 $D_e$)
= 0.015985 × 1.2041 × 7.0736² / (2 × 0.6)
= 0.80253 Pa/m
Friction rate, riser
[27] $dp_{f,s2}$ = riser: (Δp/L) L
= 0.80253 × 12
= 9.6303 Pa
Friction loss, riser

[28] $dp_{fit,s2}$ = riser: ΣC ρ v²/2
= 0.3 × 30.124
= 9.0371 Pa
Fitting loss, riser
[29] $dp_{s2}$ = riser: friction + fittings + equipment
= 9.6303 + 9.0371 + 0
= 18.667 Pa
Section loss, riser (index run)
[30] $a_{s3}$ = floor main: side a
= as given
= 500 mm
Side a, floor main
[31] $b_{s3}$ = floor main: side b
= from the target equivalent diameter 473.16 mm
= 400 mm
Side b, floor main
[32] $De_{s3}$ = floor main: equivalent diameter [1](Ch. 21)
= 1.30 (ab)^0.625/(a + b)^0.25
= 488.12 mm
Equivalent diameter, floor main
[33] $v_{s3}$ = floor main: Q / A
= 1.2 / 0.2
= 6 m/s
Velocity, floor main
[34] $Re_{s3}$ = floor main: ρ $v_e$ $D_e$ / μ ($v_e$ = 4Q/π$D_e$²)
= 1.2041 × 6.4127 × 0.48812 / 0.000018133
= 207850
Reynolds number, floor main
[35] $f_{s3}$ = floor main: friction factor
= Colebrook–White
= 0.016905
Friction factor, floor main
[36] $dpL_{s3}$ = floor main: f ρ $v_e$² / (2 $D_e$)
= 0.016905 × 1.2041 × 6.4127² / (2 × 0.48812)
= 0.85744 Pa/m
Friction rate, floor main
[37] $dp_{f,s3}$ = floor main: (Δp/L) L
= 0.85744 × 25
= 21.436 Pa
Friction loss, floor main
[38] $dp_{fit,s3}$ = floor main: ΣC ρ v²/2
= 1.1 × 21.674
= 23.841 Pa
Fitting loss, floor main
[39] $dp_{s3}$ = floor main: friction + fittings + equipment
= 21.436 + 23.841 + 0
= 45.277 Pa
Section loss, floor main (index run)
[40] $D_{s4}$ = branch to VAV: diameter
= target 312.82 mm rounded up to 50 mm
= 350 mm
Diameter, branch to VAV

[41] $v_{s4}$ = branch to VAV: Q / A
= 0.4 / 0.096211
= 4.1575 m/s
Velocity, branch to VAV
[42] $Re_{s4}$ = branch to VAV: ρ $v_e$ $D_e$ / μ ($v_e$ = 4Q/π$D_e^2$)
= 1.2041 × 4.1575 × 0.35 / 0.000018133
= 96625
Reynolds number, branch to VAV
[43] $f_{s4}$ = branch to VAV: friction factor
= Colebrook–White
= 0.01938
Friction factor, branch to VAV
[44] $dpL_{s4}$ = branch to VAV: f ρ $v_e^2$ / (2 $D_e$)
= 0.01938 × 1.2041 × 4.1575² / (2 × 0.35)
= 0.57623 Pa/m
Friction rate, branch to VAV
[45] $dp_{f,s4}$ = branch to VAV: (Δp/L) L
= 0.57623 × 10
= 5.7623 Pa
Friction loss, branch to VAV
[46] $dp_{fit,s4}$ = branch to VAV: ΣC ρ v²/2
= 0.9 × 10.406
= 9.3657 Pa
Fitting loss, branch to VAV
[47] $dp_{s4}$ = branch to VAV: friction + fittings + equipment
= 5.7623 + 9.3657 + 60
= 75.128 Pa
Section loss, branch to VAV (index run)
[48] $D_{s5}$ = other branch: diameter
= target 406.05 mm rounded up to 50 mm
= 450 mm
Diameter, other branch
[49] $v_{s5}$ = other branch: Q / A
= 0.8 / 0.15904
= 5.0301 m/s
Velocity, other branch
[50] $Re_{s5}$ = other branch: ρ $v_e$ $D_e$ / μ ($v_e$ = 4Q/π$D_e^2$)
= 1.2041 × 5.0301 × 0.45 / 0.000018133
= 150310
Reynolds number, other branch
[51] $f_{s5}$ = other branch: friction factor
= Colebrook–White
= 0.017809
Friction factor, other branch
[52] $dpL_{s5}$ = other branch: f ρ $v_e^2$ / (2 $D_e$)
= 0.017809 × 1.2041 × 5.0301² / (2 × 0.45)
= 0.60286 Pa/m
Friction rate, other branch
[53] $dp_{f,s5}$ = other branch: (Δp/L) L
= 0.60286 × 18
= 10.851 Pa
Friction loss, other branch

[54] $dp_{fit,s5}$ = other branch: ΣC ρ v²/2
= 1 × 15.233
= 15.233 Pa
Fitting loss, other branch
[55] $dp_{s5}$ = other branch: friction + fittings + equipment
= 10.851 + 15.233 + 60
= 86.084 Pa
Section loss, other branch
[56] $dp_{index}$ = Σ section losses on the index run
= 4 section(s)
= 169.04 Pa
Index-run pressure drop
[57] $dp_{all}$ = Σ section losses, all sections
= 5 section(s)
= 255.12 Pa
Sum over every section (not a path)
[58] $v_{peak}$ = highest section velocity
=
= 7.0736 m/s
Highest velocity
[59] $u_{dp}$ = $dp_{index}$ / $dp_{limit}$
= 169.04 / 250
= 0.67616
Pressure-drop utilisation
[60] $u_v$ = $v_{peak}$ / $v_{limit}$
= 7.0736 / 8
= 0.88419
Velocity utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $a_{s1}$ | 800 mm | a, AHU to riser |
| $b_{s1}$ | 400 mm | b, AHU to riser |
| $De_{s1}$ | 609.3 mm | $D_e$, AHU to riser |
| $v_{s1}$ | 6.250 m/s | v, AHU to riser |
| $Re_{s1}$ | 277500 | Re, AHU to riser |
| $f_{s1}$ | 0.015999 | f, AHU to riser |
| $dpL_{s1}$ | 0.7435 Pa/m | Δp/L, AHU to riser |
| $dp_{f,s1}$ | 11.15 Pa | Friction loss, AHU to riser |
| $dp_{fit,s1}$ | 18.81 Pa | Fitting loss, AHU to riser |
| $dp_{s1}$ | 29.97 Pa | Section loss, AHU to riser |
| $D_{s2}$ | 600 mm | D, riser |
| $v_{s2}$ | 7.074 m/s | v, riser |
| $Re_{s2}$ | 281820 | Re, riser |
| $f_{s2}$ | 0.015985 | f, riser |
| $dpL_{s2}$ | 0.8025 Pa/m | Δp/L, riser |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $dp_{f,s2}$ | 9.630 Pa | Friction loss, riser |
| $dp_{fit,s2}$ | 9.037 Pa | Fitting loss, riser |
| $dp_{s2}$ | 18.67 Pa | Section loss, riser |
| $a_{s3}$ | 500 mm | a, floor main |
| $b_{s3}$ | 400 mm | b, floor main |
| $De_{s3}$ | 488.1 mm | $D_e$, floor main |
| $v_{s3}$ | 6 m/s | v, floor main |
| $Re_{s3}$ | 207850 | Re, floor main |
| $f_{s3}$ | 0.016905 | f, floor main |
| $dpL_{s3}$ | 0.8574 Pa/m | Δp/L, floor main |
| $dp_{f,s3}$ | 21.44 Pa | Friction loss, floor main |
| $dp_{fit,s3}$ | 23.84 Pa | Fitting loss, floor main |
| $dp_{s3}$ | 45.28 Pa | Section loss, floor main |
| $D_{s4}$ | 350 mm | D, branch to VAV |
| $v_{s4}$ | 4.158 m/s | v, branch to VAV |
| $Re_{s4}$ | 96625 | Re, branch to VAV |
| $f_{s4}$ | 0.01938 | f, branch to VAV |
| $dpL_{s4}$ | 0.5762 Pa/m | Δp/L, branch to VAV |
| $dp_{f,s4}$ | 5.762 Pa | Friction loss, branch to VAV |
| $dp_{fit,s4}$ | 9.366 Pa | Fitting loss, branch to VAV |
| $dp_{s4}$ | 75.13 Pa | Section loss, branch to VAV |
| $D_{s5}$ | 450 mm | D, other branch |
| $v_{s5}$ | 5.030 m/s | v, other branch |
| $Re_{s5}$ | 150310 | Re, other branch |
| $f_{s5}$ | 0.017809 | f, other branch |
| $dpL_{s5}$ | 0.6029 Pa/m | Δp/L, other branch |
| $dp_{f,s5}$ | 10.85 Pa | Friction loss, other branch |
| $dp_{fit,s5}$ | 15.23 Pa | Fitting loss, other branch |
| $dp_{s5}$ | 86.08 Pa | Section loss, other branch |
| $dp_{index}$ | 169.0 Pa | Index-run Δp |
| $dp_{all}$ | 255.1 Pa | Σ all sections |
| $v_{peak}$ | 7.074 m/s | Highest velocity |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Velocity ≤ limit | $u_v$ = 0.88 | $u_v$ ≤ 1 | 0.88 | |
| PASS | Index-run Δp ≤ limit | $u_{dp}$ = 0.68 | $u_{dp}$ ≤ 1 | 0.68 | |

## 7. Applicability

None.

Defaults used (inputs left empty):
$v_{target}$: not entered, so the default of 6 m/s is used.

## 8. Verification

```
Template bs.duct-sizing@1.1.0 Content hash 82e9365c81cc4726 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
3. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Rectangular Cross-Section—Dimensions. BS EN 1505. BSI.
4. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Circular Cross-Section—Dimensions. BS EN 1506. BSI.